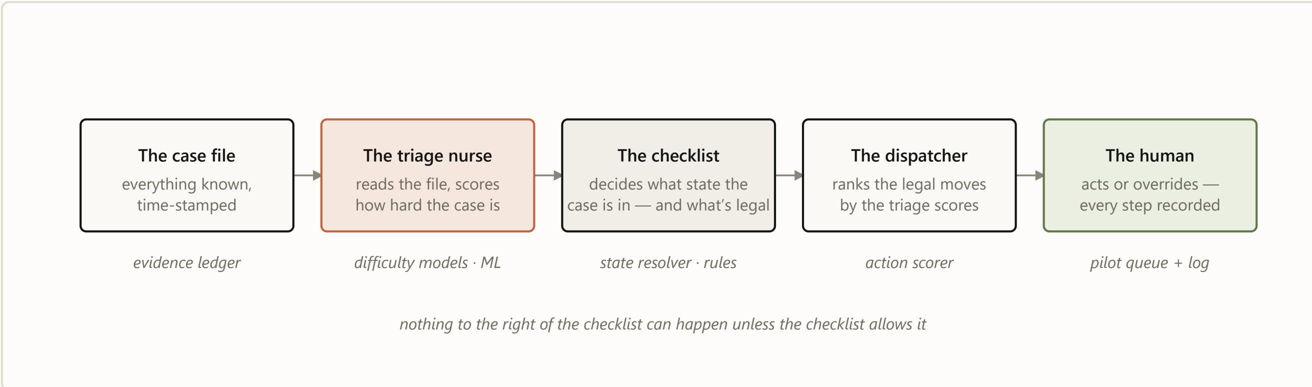
Meet **case 4127**: an employment verification, opened on a Monday. There is a work email and a phone number on file, no fax. Somewhere, an HR inbox is about to be asked to confirm a job history. Over the next nine days this case will be emailed twice, called once, and finally verified — and at every step, the system has to answer the same two questions: *how is this case doing?* and *what, if anything, should we do next?*

Five components take turns answering. Here is the cast, and the color code every diagram in this post uses:



evidence & decisions　models — probabilistic, *predicts*　rules — deterministic, *permits*

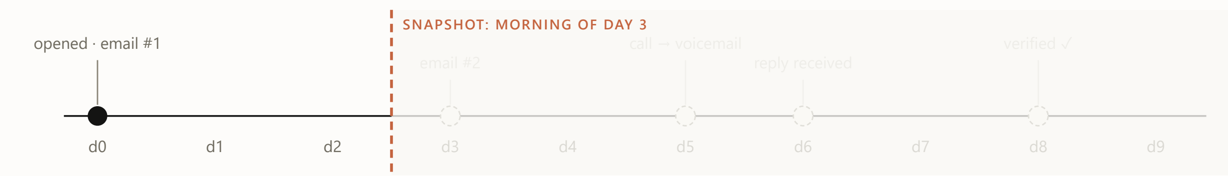
humans — decide & log

FIGURE 1 The cast, in the order they speak. The color code holds for every diagram below: clay predicts, oat permits, moss decides.

1 The curtain: what the system is allowed to know

The single most important idea in the architecture is not a model — it's a rule about time. The system never looks at a case as one blob of history. It looks at the case *as of a specific morning*, and it may only use facts from **before** that morning. One case therefore becomes many training rows — one per day it stayed open — and our twelve months of history becomes 2,751,900 of these `(case, day)` snapshots.

Drag the day below and watch what case 4127 looks like from inside the system. Everything right of the curtain hasn't happened yet — as far as the machine knows, it never will.



`always visible — intake facts: product EMPV · contacts on file: email ✓ phone ✓ fax ✗ · client policy`

Snapshot day — day 3

WHAT THE MODEL SEES (AS-OF FEATURES)	
attempts so far	1
last action	email #1 (day 0)
days since last touch	3
inbound response on record	no
note-taxonomy flags	–

WHAT IT PREDICTS · WHAT THE RULES SAY	
verifies ≤ 2 days	19%
verifies ≤ 5 days	43%
never verifies	30%

RESOLVER STATE: OUTREACH READY

Day 3. Cooldown expired, still no reply. Only now do the scores matter: the dispatcher ranks the legal moves, and email #2 goes out later today.

FIGURE 2 The as-of curtain, interactively. Probabilities are illustrative. This curtain is also the moral of our leakage story: an early model scored 0.93 because post-outcome fields let it peek right of the line. Rebuilt with strict as-of discipline, the honest number is 0.71 — and an automated audit now checks the curtain on every new training matrix.

2 Triage: three questions about the same case

Each morning, three calibrated models read the visible half of the file and score it: *will this verify within 2 days? within 5? will it ever?* Calibrated means the numbers behave like frequencies — across all cases scored "30% never-verify," about 30% actually never verify. That property is what makes the scores safe to build queues and thresholds on.

One case's scores are mildly interesting. The population view is where triage earns its keep: sorted by predicted never-verify risk, the easiest tenth of cases fails 3% of the time and the hardest tenth 48%. Same process, same operators — a 16× spread in what was ever achievable. That spread powers the queue (work the winnable middle first) and a fairer scoreboard (measure misses against what was possible, not against a blended average).

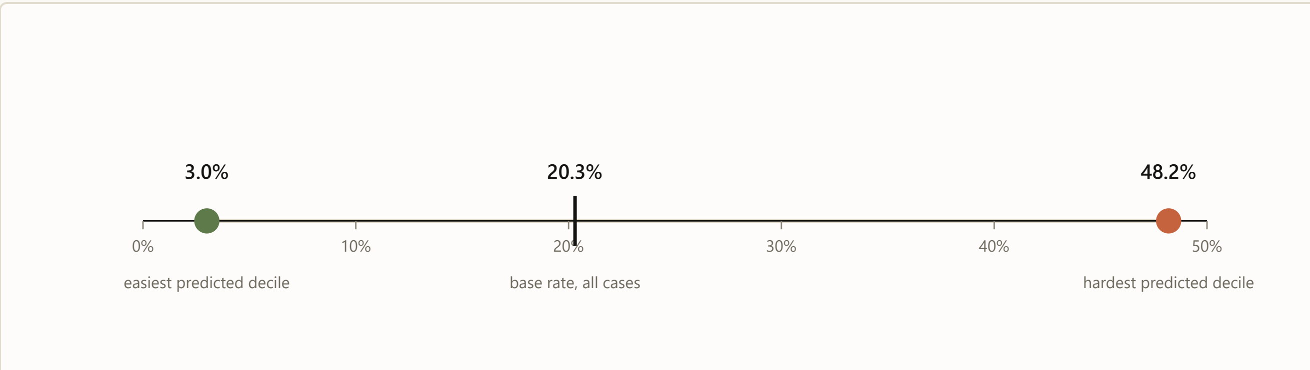


FIGURE 3 Real numbers (May 2026 holdout): observed never-verify rate by predicted-risk decile. Triage doesn't change any single case; it changes which cases get the attention.

3 The checklist: what state is this case actually in?

Before anyone acts on a score, a deterministic resolver asks a fixed sequence of questions any good supervisor would ask — and the first "yes" decides the case's state for the day. The real resolver distinguishes thirteen states; the intuition fits in seven questions:

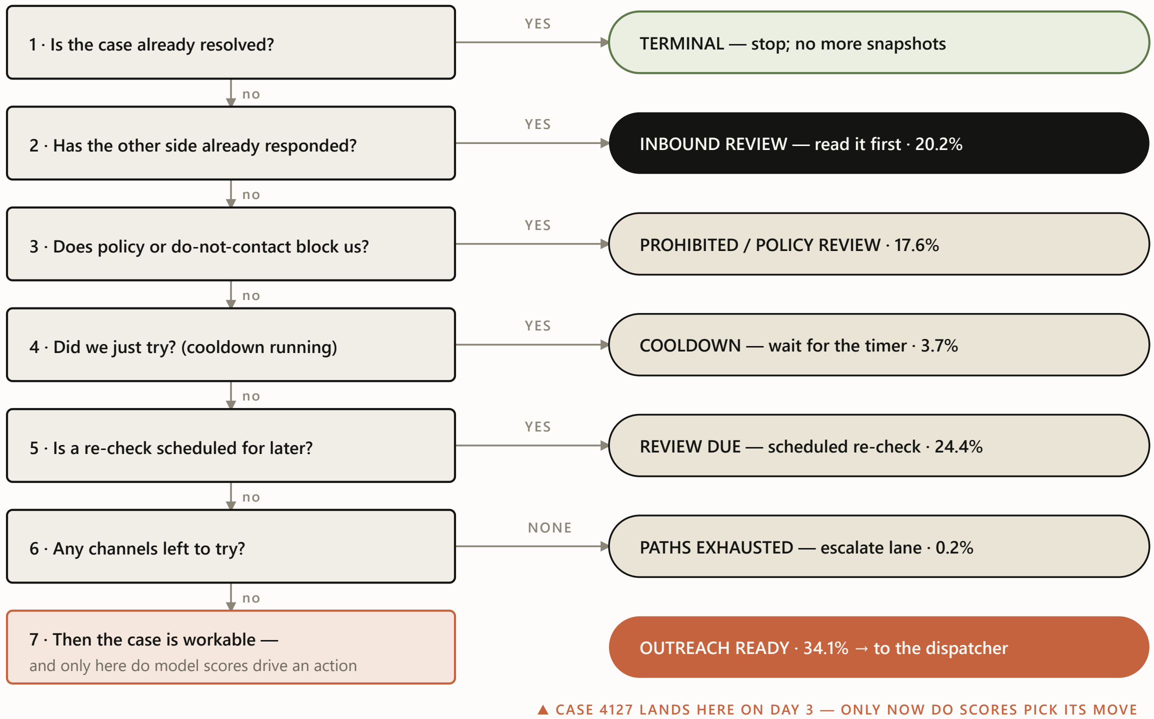


FIGURE 4 The resolver as a supervisor's checklist (a seven-question simplification of the real thirteen states; percentages are each state's share of all 2.75M case-days). Notice question 2: a fifth of all case-days already have a response on record. Answering it deterministically is what caught the ~1-in-7 cases the old flow would have re-contacted. Across all 2.75M case-days: zero do-not-contact violations, zero policy violations — by construction, since scores are never consulted until row 7.

4 The bouncer and the dispatcher: six cards, one recommendation

Suppose it's day 3 and case 4127 landed on "outreach ready." The system now deals six possible moves — the same six for every case, 16.5 million candidate cards in the full run. Eligibility rules physically remove cards before any ranking happens; the model's opinion of a blocked card is irrelevant, because the card is no longer on the table.

FIGURE 5 Case 4127's hand on day 3. Priority bars are illustrative; ranking reflects what historically moved similar cases under current policy — an observational signal, deliberately not a claim that any channel *causes* better outcomes. Blocked cards never reach the ranker: eligibility is a filter, not a penalty.

The dispatcher's output for the day is exactly one line: *recommended next move: EMAIL, rank 1 of 4 legal*. In the full backtest, every single case-group ended with an available move — zero "rank-1 but blocked" contradictions, zero empty hands.

5 The flight recorder: a person decides, everything is written down

The recommendation lands in a queue in front of a person, who can take it or override it — and the override is as valuable a signal as the action. Five event types get logged, and together they form the case's complete decision story:

day 3 · 09:12	RECOMMENDED EMAIL — rank 1 of 4 legal moves, shown to operator
day 3 · 09:14	ACTION EMAIL sent, no override · template family `follow_up` v3
day 4 · 07:40	DELIVERY delivered — no bounce
day 6 · 15:03	REPLY inbound response recorded → resolver flips to INBOUND REVIEW
day 8 · 11:26	OUTCOME verified — outcome joined back to every prior event

FIGURE 6 Case 4127's flight record (fictional timestamps). This chain — recommendation, action or override, template version, delivery and reply, outcome — is the price of ever saying the word "because." Until it's collected at scale, the system reports correlations as observations and nothing more.

6 The heartbeat: how the machine stays honest month over month

Everything above is the *daily* loop. Wrapped around it is a monthly one: new data lands, the snapshot matrix is rebuilt behind the curtain, an automated leakage audit checks the curtain held, challenger models are trained, and a mechanical gate decides — champion or challenger — with the verdict recorded either way. The gate has said no to three of our own improvements this year. That's the feature, not the bug.

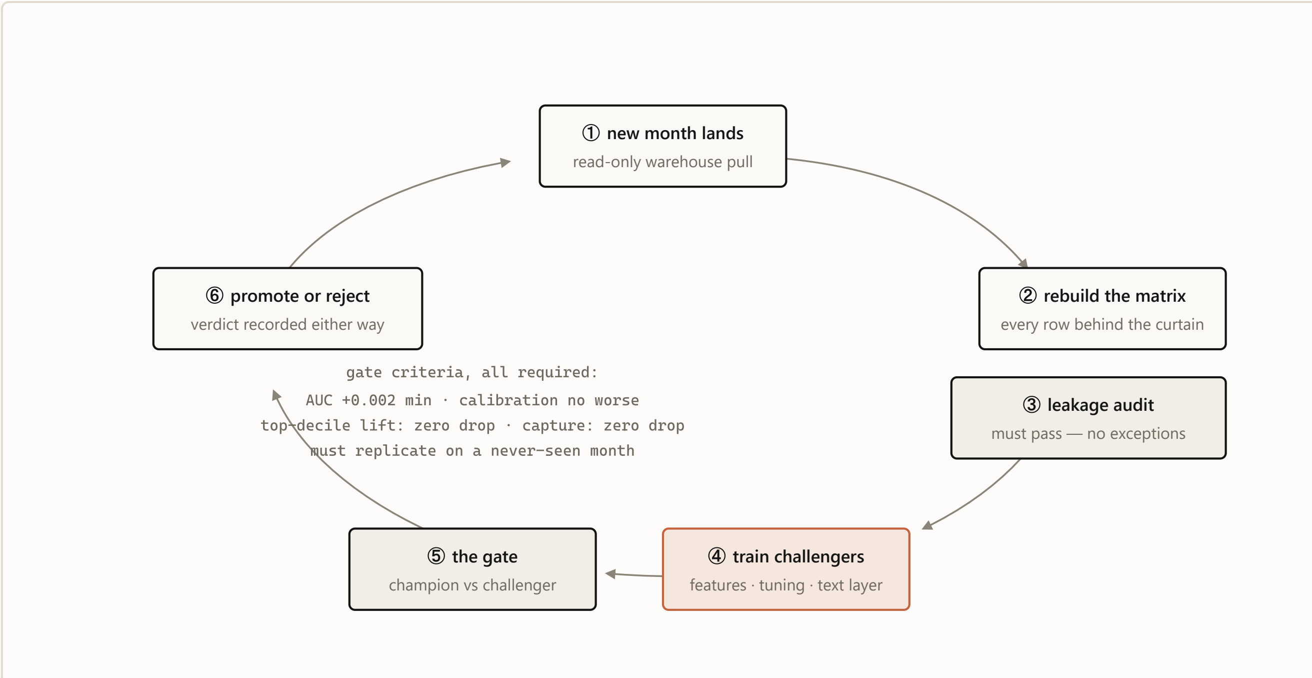


FIGURE 7 The monthly heartbeat. July 2026's cycle promoted text-aware champions for both success targets and rejected the same idea for the never-verify target — its third rejection this year, each one recorded in the champion manifest. A gate that never says no is a rubber stamp.

WHY TWO LANES AND NOT ONE BIG MODEL?

Because the two lanes fail differently. If triage is wrong, an operator works cases in a slightly worse order — cheap, recoverable, invisible. If "what's legal" is wrong, someone gets contacted who must never be contacted — an incident with a name on it. Splitting the lanes means the expensive failure mode is handled by the component you can audit line by line, and the learning component is confined to the failure mode you can afford. A single end-to-end model would blend both failure modes into one — and price everything at the expensive one.

The whole machine on one page

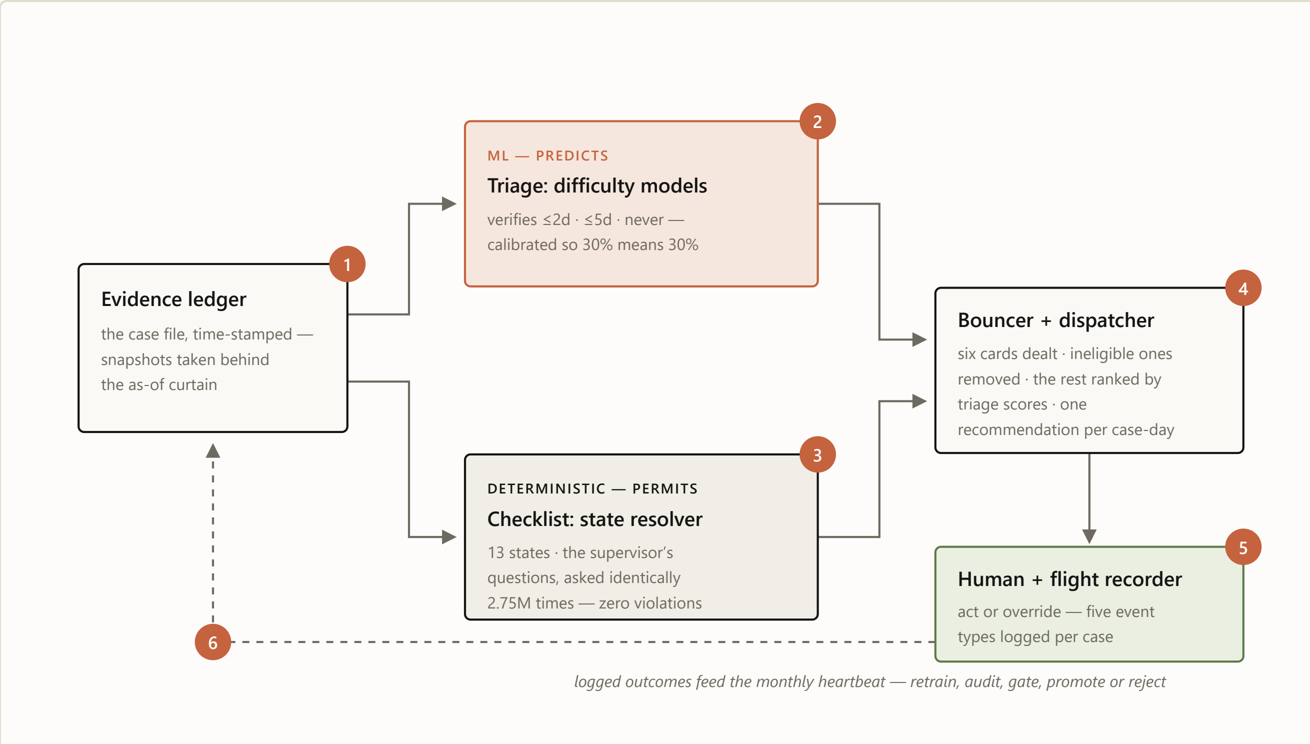


FIGURE 8 The assembled machine, with case 4127's six stops numbered in order: **1** the file is snapshotted behind the curtain · **2** triage scores it · **3** the checklist names its state · **4** the bouncer deals and the dispatcher ranks · **5** a person decides and the recorder writes it down · **6** the logs feed the monthly heartbeat. Clay predicts; oat permits; moss decides.

That's the whole machine. No step is mysterious on its own — a curtain, three scores, a checklist, a bouncer, a queue, a log — and that is deliberate. Each layer is simple enough to explain to the people whose work it touches, and the intelligence lives in how strictly the layers are allowed to talk to each other. One layer remains on the drawing board, on purpose: once the flight recorder holds enough history, a small language model will draft the communications themselves — the emails, faxes, notes, and call scripts — learning from past communication to propose each case's best next move. ◆

The series: Part 1 — Building a decision engine for verification casework (the engineering deep dive) · Part 2 — Earned autonomy (the design philosophy) · Part 3 — this illustrated guide · Plus: the executive briefing and the Murphy Brown field report.

Case 4127 is fictional and its probabilities, ranks, and timestamps are illustrative. Real numbers cited: 2,751,900 case-day snapshots; 16.5M candidate actions; state-mix percentages from the 2026-06-19 resolver run; the 3.0% / 48.2% decile spread and 0.93 → 0.71 leakage correction from the May 2026 holdout; zero DNC and policy violations across the full backtest. Correlations are observational — no channel or wording is claimed to cause outcomes. Sources in-repo: `README.md`, `presentations/ARCHITECTURE.md`, `docs/ADR/0001`, `docs/LEAKAGE_RETRACTION.md`, `config/model_champion_v1.json`.

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